

Res	Demo	Metric	Population Size Unknown				Population Size Known					
			B	WL	UA	UG	C	B	WA	WL	UG	RUT
Block	Gender	r	0.183	0.576	0.554	0.585	0.957	0.958	0.960	0.960	0.959	0.959
		R^2	0.033	0.333	0.308	0.344	0.922	0.914	0.916	0.917	0.916	0.915
	Race	r	0.195	0.480	0.462	0.487	0.620	0.708	0.701	0.700	0.704	0.714
		R^2	0.039	0.235	0.218	0.243	0.377	0.216	0.320	0.376	0.391	0.449
BG	Gender	r	0.389	0.671	0.670	0.667	0.980	0.982	0.982	0.982	0.979	0.982
		R^2	0.150	0.449	0.448	0.444	0.959	0.963	0.964	0.964	0.958	0.964
	Race	r	0.569	0.692	0.690	0.683	0.407	0.780	0.783	0.788	0.783	0.835
		R^2	0.311	0.491	0.489	0.480	0.184	0.613	0.632	0.632	0.617	0.676
Tract	Gender	r	0.458	0.723	0.726	0.723	0.985	0.986	0.987	0.987	0.982	0.988
		R^2	0.204	0.523	0.523	0.522	0.965	0.972	0.973	0.972	0.965	0.974
	Race	r	0.669	0.756	0.757	0.752	0.365	0.796	0.833	0.838	0.825	0.884
		R^2	0.457	0.585	0.587	0.578	0.151	0.640	0.705	0.709	0.696	0.761
County	Gender	r	0.809	0.986	0.985	0.988	0.999	0.999	0.999	0.999	0.999	0.999
		R^2	0.513	0.969	0.968	0.973	0.999	0.999	0.999	0.999	0.999	0.999
	Race	r	0.745	0.879	0.868	0.898	0.850	0.963	0.834	0.945	0.962	0.956
		R^2	0.491	0.740	0.726	0.775	0.694	0.826	0.551	0.837	0.886	0.796

A - Anscombe, B - (?), BG - Blockgroup, C - National Proportions, Demo - Demographic
G - Gaussian, L - Logistic, Res - Resolution, RU - Raw User, T - TFIDF, U - Normalized User, W - Normalized Word

Table 2: Performance results on multiple resolutions across gender and race/ethnicity prediction tasks. In each problem variant, bold results in a row represent configurations that are statistically indistinguishable using a paired t-test with p -value ≥ 0.05 .

find that 95% of geographic regions with at least 100 Twitter users, have low relative errors. In these regions, the relative error is at most 1.98 (block), 1.15 (blockgroup), 0.90 (tract), and 0.78 (county).

6 Conclusion

In this paper, we have shown that geotagged Tweets can be used to estimate demographics of high-resolution geographies. Our method can be used as an alternative or a complement to survey methods. We have shown that certain feature transformations such as Anscombe, TFIDF, Logistic, and Gaussian significantly improve prediction performance relative to competing baselines. We have also shown that our method is able to learn proportions of demographic categories and can provide accurate predictions at regions with at least 100 Twitter users.

For future work, it is worth bringing attention to the effect of data sampling rate on prediction. According to ? (?), word frequencies normalized by users (Normalized User) are not invariant to the sampling rate of the data. If we remove half the tweets, then these frequencies will decrease because the number of users will decrease more slowly than raw word counts. So, it would be interesting to investigate methods that can minimize the variance of such normalizations to the sampling rate.

7 Acknowledgments

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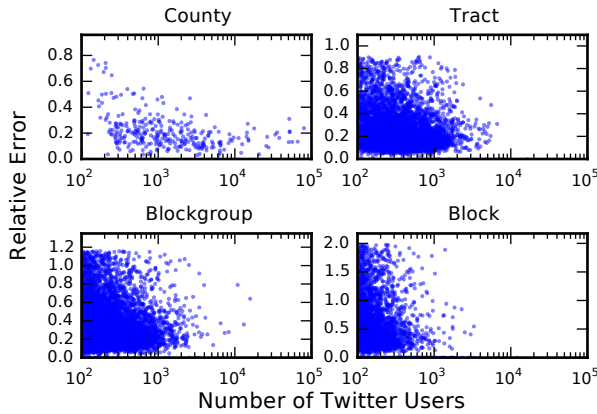


Figure 1: Relative error for 95% of geographic regions with at least 100 Twitter users (Raw User with TFIDF).

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